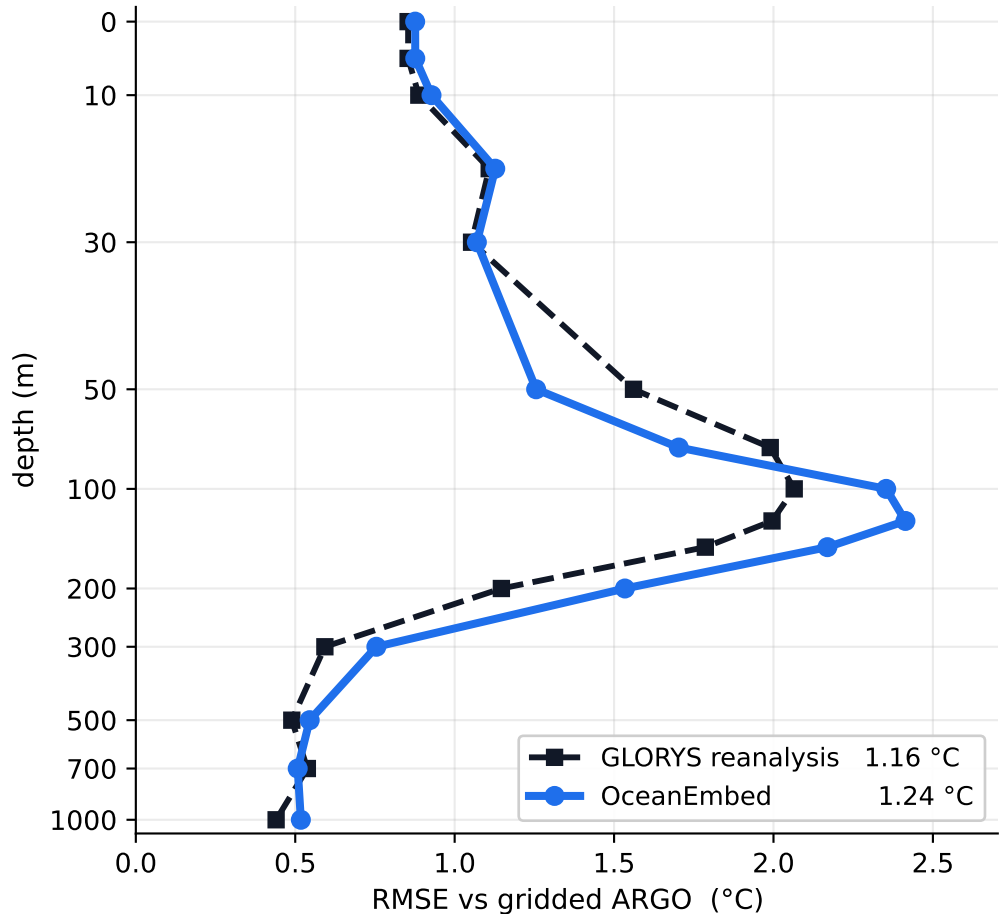


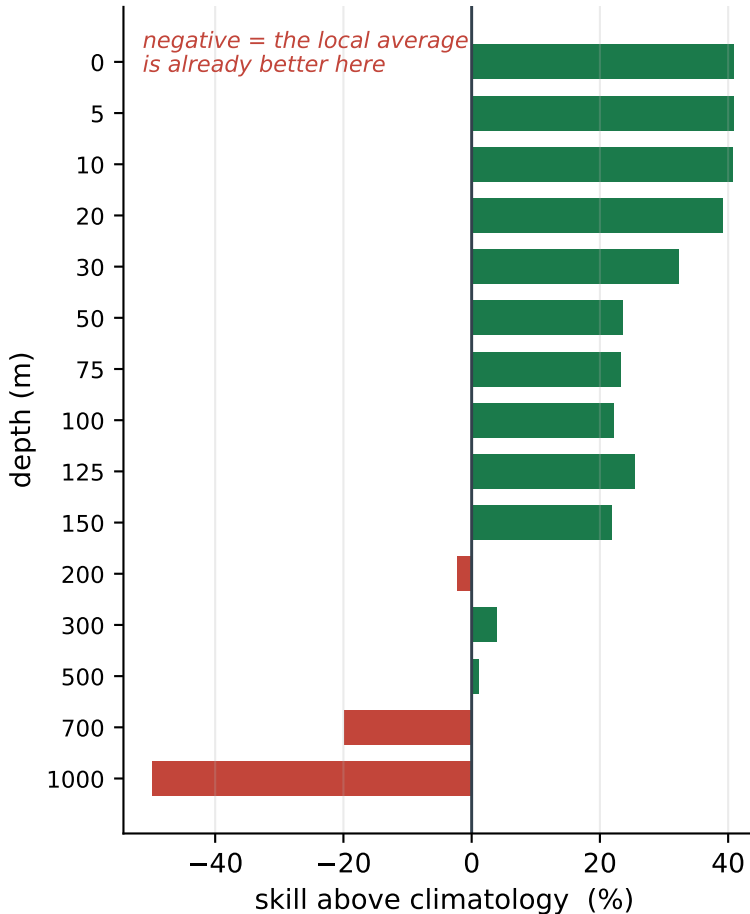
Reconstructed from satellites alone -- and checked twice

Left: against an independent float product neither model was trained on. Right: against the reanalysis, on days never seen in training.

We track the reanalysis we learn from



And we beat the do-nothing answer



7%

behind a supercomputer reanalysis, on an independent reference.

+16%

better than climatology on average across all 15 depths.

0.886

correlation, 93,600 held-out profiles.

Reference on the left is gridded ARGO (INCOIS argo_10dv), a 1° 10-day objective analysis built from float profiles -- not the raw profiles. It is smoothed, while we run at 0.25° daily, so structure that we resolve and it has smoothed away counts as our error. That makes the left panel a conservative test, and it is why climatology is not drawn there: a smooth field scores well against a smooth reference by construction. Skill above climatology is therefore measured on the right, against GLORYS on held-out days, where that comparison means something.